

Probability and Statistics

Comprehensive Exam Question Bank
10 Parallel Test Variants

Format: Multiple Choice Questions (1-10 per Variant)

Scope: Descriptive Statistics, Probability Rules, Expected Values, Normal Distribution, Confidence Intervals &
Hypothesis Testing

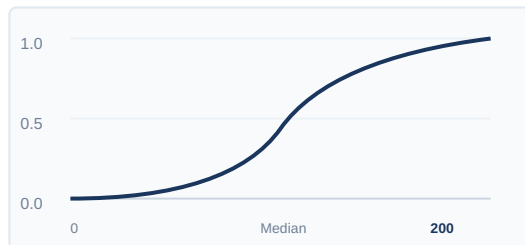
Target Audience: University Level / Academic Evaluation

Probability & Statistics — Variant 1

1. An outlier is best defined as an observation that:

- A. is seen more frequently than any other observations in the data set.
- B. is seen less frequently than the other observations in the data set.
- C. is always smaller than the rest of the observations in the data set.
- D. is always larger than the rest of the observations in the data set.
- E. is significantly and abnormally different from the other observations in the data set.

2. Management tracks the daily demand for flu shots over several years. A cumulative probability chart compiled from the data follows a standard sigmoidal path where the 95th percentile corresponds precisely to a value of 200 Shots.



How capacity or inventory should the center maintain daily to guarantee they fully satisfy demand on 95 percent of the days?

- A. At most 180 units
- B. At most 200 units
- C. Exactly 170 units
- D. At least 150 units
- E. At least 200 units

3. A large corporation has two separate primary operational divisions. The mean performance metric (or salary compensation baseline) of staff in Division 1 is 28,500 based on a cohort size of 128 employees. Division 2 features a mean baseline of 22,500 across a cohort size of 32 employees. What is the overall combined mean metric paid/allocated across all combined employees in this corporation?

- A. 22,500
- B. 24,500
- C. 25,500
- D. 27,300
- E. 28,500

4. A regional school district collects 75 years of weather closure datasets and establishes that the frequency distribution for days missed per year exhibits a pronounced right-hand skew (positive skewness) with an isolated far-right extreme outlier. Which of the following parameters should be chosen as the most accurate, robust measure to describe the center of this distribution?

- A.** Mean, because it acts as an unbiased estimator for all linear samples.
- B.** Median, because it remains resistant to extreme values and skewness.
- C.** IQR, because it eliminates outliers and accounts for the central 50 percent.
- D.** First quartile, because the distribution is left-skewed.
- E.** Standard deviation, because it represents structural spread without bias.

5. The probability of an operational infrastructure failure incident occurring on any given business day depends strongly on environmental conditions. If weather systems remain dry, there is a 0.2% chance of an incident. If weather conditions shift to wet, the probability escalates to 1.0%. Today, the local meteorological station forecasts a 20% probability of wet conditions. What is the total probability that an operational incident occurs today?

- A.** 0.0004
- B.** 0.0016
- C.** 0.0020
- D.** 0.0036
- E.** 0.0060

6. An institutional recruitment agency tracks retention rates carefully. Historical logs show that precisely 20% of all placed contractual candidates leave their positions by the end of their first year, and only 5% of those who successfully complete the first year leave during the subsequent second year. If an employer onboarded 100 workers via this firm at the start of Year 1, and onboarded another 100 workers at the start of Year 2, how many total workers are statistically expected to remain active on the job at the exact end of the second year?

- A.** 140
- B.** 148
- C.** 152
- D.** 156
- E.** 163

7. The nutritional caloric value of a specific commercial item is normally distributed with an expected mean of 210 calories and a standard deviation of 10 calories. What percentage of these items contain more than 225 calories?

- A. 66.8%**
- B. 47.7%**
- C. 43.3%**
- D. 6.68%**
- E. 3.34%**

8. A global distributor utilizes standard protective shipping units. The internal product contents packed per unit exhibit a mean weight of 25 lbs with a standard deviation of 2.0 lbs. The physical container itself has a standard deviation of 0.15 lbs, and the internal stabilization material has a standard deviation of 0.25 lbs. Assuming all component weights are independent, what is the exact standard deviation of the total packed shipping units?

- A. 28.000 pounds**
- B. 5.290 pounds**
- C. 4.085 pounds**
- D. 2.400 pounds**
- E. 2.021 pounds**

9. A lead social researcher wants to estimate the true population mean annual income per family within a metropolitan school zone. Following a rigorous simple random sample, the researcher establishes a 95 percent confidence interval for the population mean spanning from \$35,095 to \$45,005. Which statement provides the most accurate statistical interpretation of this confidence interval?

- A. 95 percent of all families in this school zone have an annual income that falls within the interval limits.**
- B. There is a 95% probability that the families of all students combined earn exactly between \$35,095 and \$45,005.**
- C. If we were to take another sample of identical size and calculate a 95% confidence interval, there is a 95% chance that the new interval bounds will exactly match (\$35,095, \$45,005).**
- D. There is a 95% probability that the mean of a subsequent sample drawn with matching sample size will fall between \$35,095 and \$45,005.**
- E. In long-run repeated sampling, 95% of intervals generated using this exact methodology will successfully encompass the true fixed population mean income.**

10. The municipal Department of Environmental Health tests soil toxicity across a local reserve. If the contamination level exceeds regulatory thresholds, the reserve is ordered to close immediately. The null hypothesis states H_0 : "The reserve is safe and non-toxic". Which option describes a final operational decision resulting directly from a Type I error?

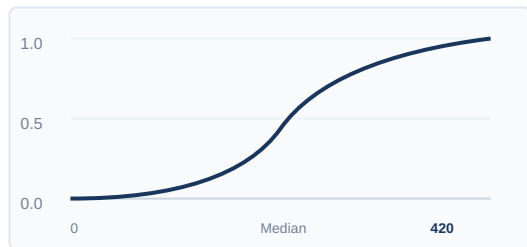
- A.** Closing down the reserve when the toxicity levels are actually within regulatory safety limits (false alarm).
- B.** Keeping the reserve open and operating when toxicity levels are dangerously in excess of limits.
- C.** Closing down the reserve when toxicity levels are genuinely in excess of allowed limits.
- D.** Keeping the reserve open and operating when toxicity levels are completely safe and within limits.
- E.** Closing the reserve entirely due to unrelated infrastructure noise level constraints.

Probability & Statistics — Variant 2

1. An outlier is best defined as an observation that:

- A. is seen more frequently than any other observations in the data set.
- B. is seen less frequently than the other observations in the data set.
- C. is always smaller than the rest of the observations in the data set.
- D. is always larger than the rest of the observations in the data set.
- E. is significantly and abnormally different from the other observations in the data set.

2. Management tracks the daily demand for electricity demand (MW) over several years. A cumulative probability chart compiled from the data follows a standard sigmoidal path where the 95th percentile corresponds precisely to a value of 420 MW.



How capacity or inventory should the center maintain daily to guarantee they fully satisfy demand on 95 percent of the days?

- A. At most 378 units
- B. At most 420 units
- C. Exactly 357 units
- D. At least 315 units
- E. At least 420 units

3. A large corporation has two separate primary operational divisions. The mean performance metric (or salary compensation baseline) of staff in Division 1 is 45 based on a cohort size of 60 employees. Division 2 features a mean baseline of 55 across a cohort size of 40 employees. What is the overall combined mean metric paid/allocated across all combined employees in this corporation?

- A. 55
- B. -950
- C. 50
- D. 49.0
- E. 45

4. A regional school district collects 75 years of weather closure datasets and establishes that the frequency distribution for days missed per year exhibits a pronounced right-hand skew (positive skewness) with an isolated far-right extreme outlier. Which of the following parameters should be chosen as the most accurate, robust measure to describe the center of this distribution?

- A.** Mean, because it acts as an unbiased estimator for all linear samples.
- B.** Median, because it remains resistant to extreme values and skewness.
- C.** IQR, because it eliminates outliers and accounts for the central 50 percent.
- D.** First quartile, because the distribution is left-skewed.
- E.** Standard deviation, because it represents structural spread without bias.

5. The probability of an operational infrastructure failure incident occurring on any given business day depends strongly on environmental conditions. If weather systems remain dry, there is a 1.0% chance of an incident. If weather conditions shift to wet, the probability escalates to 5.0%. Today, the local meteorological station forecasts a 30% probability of wet conditions. What is the total probability that an operational incident occurs today?

- A.** 0.0004
- B.** 0.0016
- C.** 0.0020
- D.** 0.0220
- E.** 0.0060

6. An institutional recruitment agency tracks retention rates carefully. Historical logs show that precisely 10% of all placed contractual candidates leave their positions by the end of their first year, and only 5% of those who successfully complete the first year leave during the subsequent second year. If an employer onboarded 200 workers via this firm at the start of Year 1, and onboarded another 200 workers at the start of Year 2, how many total workers are statistically expected to remain active on the job at the exact end of the second year?

- A.** 315
- B.** 333
- C.** 343
- D.** 351
- E.** 368

7. The nutritional caloric value of a specific commercial item is normally distributed with an expected mean of 100 calories and a standard deviation of 10 calories. What percentage of these items contain more than 115 calories?

- A.** 66.8%
- B.** 47.7%
- C.** 43.3%
- D.** 6.68%
- E.** 3.34%

8. A global distributor utilizes standard protective shipping units. The internal product contents packed per unit exhibit a mean weight of 25 lbs with a standard deviation of 3.0 lbs. The physical container itself has a standard deviation of 1.0 lbs, and the internal stabilization material has a standard deviation of 2.0 lbs. Assuming all component weights are independent, what is the exact standard deviation of the total packed shipping units?

- A.** 28.000 pounds
- B.** 5.290 pounds
- C.** 4.085 pounds
- D.** 2.400 pounds
- E.** 3.742 pounds

9. A lead social researcher wants to estimate the true population mean annual income per family within a metropolitan school zone. Following a rigorous simple random sample, the researcher establishes a 95 percent confidence interval for the population mean spanning from \$35,095 to \$45,005. Which statement provides the most accurate statistical interpretation of this confidence interval?

- A.** 95 percent of all families in this school zone have an annual income that falls within the interval limits.
- B.** There is a 95% probability that the families of all students combined earn exactly between \$35,095 and \$45,005.
- C.** If we were to take another sample of identical size and calculate a 95% confidence interval, there is a 95% chance that the new interval bounds will exactly match (\$35,095, \$45,005).
- D.** There is a 95% probability that the mean of a subsequent sample drawn with matching sample size will fall between \$35,095 and \$45,005.
- E.** In long-run repeated sampling, 95% of intervals generated using this exact methodology will successfully encompass the true fixed population mean income.

10. The municipal Department of Environmental Health tests soil toxicity across a local reserve. If the contamination level exceeds regulatory thresholds, the reserve is ordered to close immediately. The null hypothesis states H_0 : "The reserve is safe and non-toxic". Which option describes a final operational decision resulting directly from a Type I error?

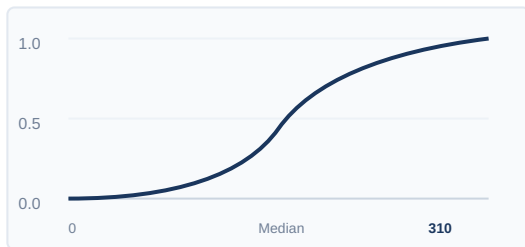
- A.** Closing down the reserve when the toxicity levels are actually within regulatory safety limits (false alarm).
- B.** Keeping the reserve open and operating when toxicity levels are dangerously in excess of limits.
- C.** Closing down the reserve when toxicity levels are genuinely in excess of allowed limits.
- D.** Keeping the reserve open and operating when toxicity levels are completely safe and within limits.
- E.** Closing the reserve entirely due to unrelated infrastructure noise level constraints.

Probability & Statistics — Variant 3

1. An outlier is best defined as an observation that:

- A. is seen more frequently than any other observations in the data set.
- B. is seen less frequently than the other observations in the data set.
- C. is always smaller than the rest of the observations in the data set.
- D. is always larger than the rest of the observations in the data set.
- E. is significantly and abnormally different from the other observations in the data set.

2. Management tracks the daily demand for water consumption (kL) over several years. A cumulative probability chart compiled from the data follows a standard sigmoidal path where the 95th percentile corresponds precisely to a value of 310 kL.



How capacity or inventory should the center maintain daily to guarantee they fully satisfy demand on 95 percent of the days?

- A. At most 279 units
- B. At most 310 units
- C. Exactly 263 units
- D. At least 232 units
- E. At least 310 units

3. A large corporation has two separate primary operational divisions. The mean performance metric (or salary compensation baseline) of staff in Division 1 is 72 based on a cohort size of 150 employees. Division 2 features a mean baseline of 88 across a cohort size of 50 employees. What is the overall combined mean metric paid/allocated across all combined employees in this corporation?

- A. 88
- B. -920
- C. 80
- D. 76.0
- E. 72

4. A regional school district collects 75 years of weather closure datasets and establishes that the frequency distribution for days missed per year exhibits a pronounced right-hand skew (positive skewness) with an isolated far-right extreme outlier. Which of the following parameters should be chosen as the most accurate, robust measure to describe the center of this distribution?

- A. Mean, because it acts as an unbiased estimator for all linear samples.
- B. Median, because it remains resistant to extreme values and skewness.
- C. IQR, because it eliminates outliers and accounts for the central 50 percent.
- D. First quartile, because the distribution is left-skewed.
- E. Standard deviation, because it represents structural spread without bias.

5. The probability of an operational infrastructure failure incident occurring on any given business day depends strongly on environmental conditions. If weather systems remain dry, there is a 5.0% chance of an incident. If weather conditions shift to wet, the probability escalates to 2.0%. Today, the local meteorological station forecasts a 40% probability of wet conditions. What is the total probability that an operational incident occurs today?

- A. 0.0004
- B. 0.0016
- C. 0.0020
- D. 0.0380
- E. 0.0060

6. An institutional recruitment agency tracks retention rates carefully. Historical logs show that precisely 15% of all placed contractual candidates leave their positions by the end of their first year, and only 8% of those who successfully complete the first year leave during the subsequent second year. If an employer onboarded 500 workers via this firm at the start of Year 1, and onboarded another 500 workers at the start of Year 2, how many total workers are statistically expected to remain active on the job at the exact end of the second year?

- A. 756
- B. 798
- C. 824
- D. 841
- E. 883

7. The nutritional caloric value of a specific commercial item is normally distributed with an expected mean of 50 calories and a standard deviation of 4 calories. What percentage of these items contain more than 56 calories?

- A. 66.8%
- B. 47.7%
- C. 43.3%
- D. 6.68%
- E. 3.34%

8. A global distributor utilizes standard protective shipping units. The internal product contents packed per unit exhibit a mean weight of 25 lbs with a standard deviation of 0.4 lbs. The physical container itself has a standard deviation of 0.2 lbs, and the internal stabilization material has a standard deviation of 0.1 lbs. Assuming all component weights are independent, what is the exact standard deviation of the total packed shipping units?

- A. 28.000 pounds
- B. 5.290 pounds
- C. 4.085 pounds
- D. 2.400 pounds
- E. 0.458 pounds

9. A lead social researcher wants to estimate the true population mean annual income per family within a metropolitan school zone. Following a rigorous simple random sample, the researcher establishes a 95 percent confidence interval for the population mean spanning from \$35,095 to \$45,005. Which statement provides the most accurate statistical interpretation of this confidence interval?

- A. 95 percent of all families in this school zone have an annual income that falls within the interval limits.
- B. There is a 95% probability that the families of all students combined earn exactly between \$35,095 and \$45,005.
- C. If we were to take another sample of identical size and calculate a 95% confidence interval, there is a 95% chance that the new interval bounds will exactly match (\$35,095, \$45,005).
- D. There is a 95% probability that the mean of a subsequent sample drawn with matching sample size will fall between \$35,095 and \$45,005.
- E. In long-run repeated sampling, 95% of intervals generated using this exact methodology will successfully encompass the true fixed population mean income.

10. The municipal Department of Environmental Health tests soil toxicity across a local reserve. If the contamination level exceeds regulatory thresholds, the reserve is ordered to close immediately. The null hypothesis states H_0 : "The reserve is safe and non-toxic". Which option describes a final operational decision resulting directly from a Type I error?

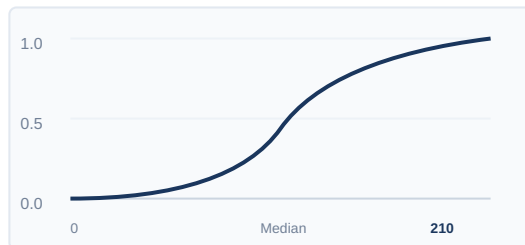
- A. Closing down the reserve when the toxicity levels are actually within regulatory safety limits (false alarm).
- B. Keeping the reserve open and operating when toxicity levels are dangerously in excess of limits.
- C. Closing down the reserve when toxicity levels are genuinely in excess of allowed limits.
- D. Keeping the reserve open and operating when toxicity levels are completely safe and within limits.
- E. Closing the reserve entirely due to unrelated infrastructure noise level constraints.

Probability & Statistics — Variant 4

1. An outlier is best defined as an observation that:

- A. is seen more frequently than any other observations in the data set.
- B. is seen less frequently than the other observations in the data set.
- C. is always smaller than the rest of the observations in the data set.
- D. is always larger than the rest of the observations in the data set.
- E. is significantly and abnormally different from the other observations in the data set.

2. Management tracks the daily demand for hotel room bookings over several years. A cumulative probability chart compiled from the data follows a standard sigmoidal path where the 95th percentile corresponds precisely to a value of 210 Rooms.



How capacity or inventory should the center maintain daily to guarantee they fully satisfy demand on 95 percent of the days?

- A. At most 189 units
- B. At most 210 units
- C. Exactly 178 units
- D. At least 157 units
- E. At least 210 units

3. A large corporation has two separate primary operational divisions. The mean performance metric (or salary compensation baseline) of staff in Division 1 is 34 based on a cohort size of 200 employees. Division 2 features a mean baseline of 44 across a cohort size of 50 employees. What is the overall combined mean metric paid/allocated across all combined employees in this corporation?

- A. 44
- B. -961
- C. 39
- D. 36.0
- E. 34

4. A regional school district collects 75 years of weather closure datasets and establishes that the frequency distribution for days missed per year exhibits a pronounced right-hand skew (positive skewness) with an isolated far-right extreme outlier. Which of the following parameters should be chosen as the most accurate, robust measure to describe the center of this distribution?

- A. Mean, because it acts as an unbiased estimator for all linear samples.
- B. Median, because it remains resistant to extreme values and skewness.
- C. IQR, because it eliminates outliers and accounts for the central 50 percent.
- D. First quartile, because the distribution is left-skewed.
- E. Standard deviation, because it represents structural spread without bias.

5. The probability of an operational infrastructure failure incident occurring on any given business day depends strongly on environmental conditions. If weather systems remain dry, there is a 2.0% chance of an incident. If weather conditions shift to wet, the probability escalates to 8.0%. Today, the local meteorological station forecasts a 10% probability of wet conditions. What is the total probability that an operational incident occurs today?

- A. 0.0004
- B. 0.0016
- C. 0.0020
- D. 0.0260
- E. 0.0060

6. An institutional recruitment agency tracks retention rates carefully. Historical logs show that precisely 20% of all placed contractual candidates leave their positions by the end of their first year, and only 10% of those who successfully complete the first year leave during the subsequent second year. If an employer onboarded 1000 workers via this firm at the start of Year 1, and onboarded another 1000 workers at the start of Year 2, how many total workers are statistically expected to remain active on the job at the exact end of the second year?

- A. 1368
- B. 1444
- C. 1489
- D. 1520
- E. 1596

7. The nutritional caloric value of a specific commercial item is normally distributed with an expected mean of 80 calories and a standard deviation of 6 calories. What percentage of these items contain more than 89 calories?

- A. 66.8%
- B. 47.7%
- C. 43.3%
- D. 6.68%
- E. 3.34%

8. A global distributor utilizes standard protective shipping units. The internal product contents packed per unit exhibit a mean weight of 25 lbs with a standard deviation of 5.0 lbs. The physical container itself has a standard deviation of 3.0 lbs, and the internal stabilization material has a standard deviation of 2.0 lbs. Assuming all component weights are independent, what is the exact standard deviation of the total packed shipping units?

- A. 28.000 pounds
- B. 5.290 pounds
- C. 4.085 pounds
- D. 2.400 pounds
- E. 6.164 pounds

9. A lead social researcher wants to estimate the true population mean annual income per family within a metropolitan school zone. Following a rigorous simple random sample, the researcher establishes a 95 percent confidence interval for the population mean spanning from \$35,095 to \$45,005. Which statement provides the most accurate statistical interpretation of this confidence interval?

- A. 95 percent of all families in this school zone have an annual income that falls within the interval limits.
- B. There is a 95% probability that the families of all students combined earn exactly between \$35,095 and \$45,005.
- C. If we were to take another sample of identical size and calculate a 95% confidence interval, there is a 95% chance that the new interval bounds will exactly match (\$35,095, \$45,005).
- D. There is a 95% probability that the mean of a subsequent sample drawn with matching sample size will fall between \$35,095 and \$45,005.
- E. In long-run repeated sampling, 95% of intervals generated using this exact methodology will successfully encompass the true fixed population mean income.

10. The municipal Department of Environmental Health tests soil toxicity across a local reserve. If the contamination level exceeds regulatory thresholds, the reserve is ordered to close immediately. The null hypothesis states H_0 : "The reserve is safe and non-toxic". Which option describes a final operational decision resulting directly from a Type I error?

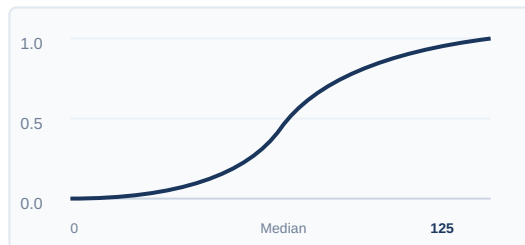
- A. Closing down the reserve when the toxicity levels are actually within regulatory safety limits (false alarm).
- B. Keeping the reserve open and operating when toxicity levels are dangerously in excess of limits.
- C. Closing down the reserve when toxicity levels are genuinely in excess of allowed limits.
- D. Keeping the reserve open and operating when toxicity levels are completely safe and within limits.
- E. Closing the reserve entirely due to unrelated infrastructure noise level constraints.

Probability & Statistics — Variant 5

1. An outlier is best defined as an observation that:

- A. is seen more frequently than any other observations in the data set.
- B. is seen less frequently than the other observations in the data set.
- C. is always smaller than the rest of the observations in the data set.
- D. is always larger than the rest of the observations in the data set.
- E. is significantly and abnormally different from the other observations in the data set.

2. Management tracks the daily demand for tech support tickets over several years. A cumulative probability chart compiled from the data follows a standard sigmoidal path where the 95th percentile corresponds precisely to a value of 125 Tickets.



How capacity or inventory should the center maintain daily to guarantee they fully satisfy demand on 95 percent of the days?

- A. At most 112 units
- B. At most 125 units
- C. Exactly 106 units
- D. At least 93 units
- E. At least 125 units

3. A large corporation has two separate primary operational divisions. The mean performance metric (or salary compensation baseline) of staff in Division 1 is 12 based on a cohort size of 70 employees. Division 2 features a mean baseline of 20 across a cohort size of 30 employees. What is the overall combined mean metric paid/allocated across all combined employees in this corporation?

- A. 20
- B. -984
- C. 16
- D. 14.4
- E. 12

4. A regional school district collects 75 years of weather closure datasets and establishes that the frequency distribution for days missed per year exhibits a pronounced right-hand skew (positive skewness) with an isolated far-right extreme outlier. Which of the following parameters should be chosen as the most accurate, robust measure to describe the center of this distribution?

- A.** Mean, because it acts as an unbiased estimator for all linear samples.
- B.** Median, because it remains resistant to extreme values and skewness.
- C.** IQR, because it eliminates outliers and accounts for the central 50 percent.
- D.** First quartile, because the distribution is left-skewed.
- E.** Standard deviation, because it represents structural spread without bias.

5. The probability of an operational infrastructure failure incident occurring on any given business day depends strongly on environmental conditions. If weather systems remain dry, there is a 1.0% chance of an incident. If weather conditions shift to wet, the probability escalates to 4.0%. Today, the local meteorological station forecasts a 25% probability of wet conditions. What is the total probability that an operational incident occurs today?

- A.** 0.0004
- B.** 0.0016
- C.** 0.0020
- D.** 0.0175
- E.** 0.0060

6. An institutional recruitment agency tracks retention rates carefully. Historical logs show that precisely 5% of all placed contractual candidates leave their positions by the end of their first year, and only 2% of those who successfully complete the first year leave during the subsequent second year. If an employer onboarded 1000 workers via this firm at the start of Year 1, and onboarded another 1000 workers at the start of Year 2, how many total workers are statistically expected to remain active on the job at the exact end of the second year?

- A.** 1692
- B.** 1786
- C.** 1843
- D.** 1881
- E.** 1975

7. The nutritional caloric value of a specific commercial item is normally distributed with an expected mean of 120 calories and a standard deviation of 12 calories. What percentage of these items contain more than 138 calories?

- A. 66.8%**
- B. 47.7%**
- C. 43.3%**
- D. 6.68%**
- E. 3.34%**

8. A global distributor utilizes standard protective shipping units. The internal product contents packed per unit exhibit a mean weight of 25 lbs with a standard deviation of 0.3 lbs. The physical container itself has a standard deviation of 0.5 lbs, and the internal stabilization material has a standard deviation of 0.2 lbs. Assuming all component weights are independent, what is the exact standard deviation of the total packed shipping units?

- A. 28.000 pounds**
- B. 5.290 pounds**
- C. 4.085 pounds**
- D. 2.400 pounds**
- E. 0.616 pounds**

9. A lead social researcher wants to estimate the true population mean annual income per family within a metropolitan school zone. Following a rigorous simple random sample, the researcher establishes a 95 percent confidence interval for the population mean spanning from \$35,095 to \$45,005. Which statement provides the most accurate statistical interpretation of this confidence interval?

- A. 95 percent of all families in this school zone have an annual income that falls within the interval limits.**
- B. There is a 95% probability that the families of all students combined earn exactly between \$35,095 and \$45,005.**
- C. If we were to take another sample of identical size and calculate a 95% confidence interval, there is a 95% chance that the new interval bounds will exactly match (\$35,095, \$45,005).**
- D. There is a 95% probability that the mean of a subsequent sample drawn with matching sample size will fall between \$35,095 and \$45,005.**
- E. In long-run repeated sampling, 95% of intervals generated using this exact methodology will successfully encompass the true fixed population mean income.**

10. The municipal Department of Environmental Health tests soil toxicity across a local reserve. If the contamination level exceeds regulatory thresholds, the reserve is ordered to close immediately. The null hypothesis states H_0 : "The reserve is safe and non-toxic". Which option describes a final operational decision resulting directly from a Type I error?

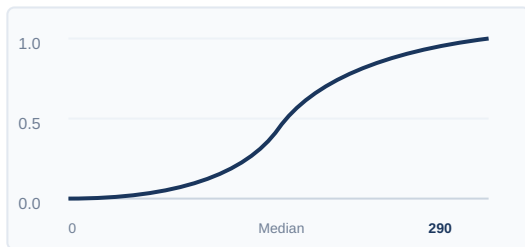
- A.** Closing down the reserve when the toxicity levels are actually within regulatory safety limits (false alarm).
- B.** Keeping the reserve open and operating when toxicity levels are dangerously in excess of limits.
- C.** Closing down the reserve when toxicity levels are genuinely in excess of allowed limits.
- D.** Keeping the reserve open and operating when toxicity levels are completely safe and within limits.
- E.** Closing the reserve entirely due to unrelated infrastructure noise level constraints.

Probability & Statistics — Variant 6

1. An outlier is best defined as an observation that:

- A. is seen more frequently than any other observations in the data set.
- B. is seen less frequently than the other observations in the data set.
- C. is always smaller than the rest of the observations in the data set.
- D. is always larger than the rest of the observations in the data set.
- E. is significantly and abnormally different from the other observations in the data set.

2. Management tracks the daily demand for daily bakery loaf sales over several years. A cumulative probability chart compiled from the data follows a standard sigmoidal path where the 95th percentile corresponds precisely to a value of 290 Loaves.



How capacity or inventory should the center maintain daily to guarantee they fully satisfy demand on 95 percent of the days?

- A. At most 261 units
- B. At most 290 units
- C. Exactly 246 units
- D. At least 217 units
- E. At least 290 units

3. A large corporation has two separate primary operational divisions. The mean performance metric (or salary compensation baseline) of staff in Division 1 is 80 based on a cohort size of 40 employees. Division 2 features a mean baseline of 40 across a cohort size of 10 employees. What is the overall combined mean metric paid/allocated across all combined employees in this corporation?

- A. 40
- B. -940
- C. 60
- D. 72.0
- E. 80

4. A regional school district collects 75 years of weather closure datasets and establishes that the frequency distribution for days missed per year exhibits a pronounced right-hand skew (positive skewness) with an isolated far-right extreme outlier. Which of the following parameters should be chosen as the most accurate, robust measure to describe the center of this distribution?

- A.** Mean, because it acts as an unbiased estimator for all linear samples.
- B.** Median, because it remains resistant to extreme values and skewness.
- C.** IQR, because it eliminates outliers and accounts for the central 50 percent.
- D.** First quartile, because the distribution is left-skewed.
- E.** Standard deviation, because it represents structural spread without bias.

5. The probability of an operational infrastructure failure incident occurring on any given business day depends strongly on environmental conditions. If weather systems remain dry, there is a 2.0% chance of an incident. If weather conditions shift to wet, the probability escalates to 6.0%. Today, the local meteorological station forecasts a 15% probability of wet conditions. What is the total probability that an operational incident occurs today?

- A.** 0.0004
- B.** 0.0016
- C.** 0.0020
- D.** 0.0260
- E.** 0.0060

6. An institutional recruitment agency tracks retention rates carefully. Historical logs show that precisely 25% of all placed contractual candidates leave their positions by the end of their first year, and only 10% of those who successfully complete the first year leave during the subsequent second year. If an employer onboarded 400 workers via this firm at the start of Year 1, and onboarded another 400 workers at the start of Year 2, how many total workers are statistically expected to remain active on the job at the exact end of the second year?

- A.** 513
- B.** 541
- C.** 558
- D.** 570
- E.** 598

7. The nutritional caloric value of a specific commercial item is normally distributed with an expected mean of 400 calories and a standard deviation of 30 calories. What percentage of these items contain more than 445 calories?

- A.** 66.8%
- B.** 47.7%
- C.** 43.3%
- D.** 6.68%
- E.** 3.34%

8. A global distributor utilizes standard protective shipping units. The internal product contents packed per unit exhibit a mean weight of 25 lbs with a standard deviation of 1.2 lbs. The physical container itself has a standard deviation of 2.0 lbs, and the internal stabilization material has a standard deviation of 0.5 lbs. Assuming all component weights are independent, what is the exact standard deviation of the total packed shipping units?

- A.** 28.000 pounds
- B.** 5.290 pounds
- C.** 4.085 pounds
- D.** 2.400 pounds
- E.** 2.385 pounds

9. A lead social researcher wants to estimate the true population mean annual income per family within a metropolitan school zone. Following a rigorous simple random sample, the researcher establishes a 95 percent confidence interval for the population mean spanning from \$35,095 to \$45,005. Which statement provides the most accurate statistical interpretation of this confidence interval?

- A.** 95 percent of all families in this school zone have an annual income that falls within the interval limits.
- B.** There is a 95% probability that the families of all students combined earn exactly between \$35,095 and \$45,005.
- C.** If we were to take another sample of identical size and calculate a 95% confidence interval, there is a 95% chance that the new interval bounds will exactly match (\$35,095, \$45,005).
- D.** There is a 95% probability that the mean of a subsequent sample drawn with matching sample size will fall between \$35,095 and \$45,005.
- E.** In long-run repeated sampling, 95% of intervals generated using this exact methodology will successfully encompass the true fixed population mean income.

10. The municipal Department of Environmental Health tests soil toxicity across a local reserve. If the contamination level exceeds regulatory thresholds, the reserve is ordered to close immediately. The null hypothesis states H_0 : "The reserve is safe and non-toxic". Which option describes a final operational decision resulting directly from a Type I error?

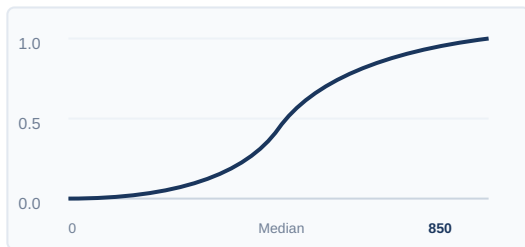
- A.** Closing down the reserve when the toxicity levels are actually within regulatory safety limits (false alarm).
- B.** Keeping the reserve open and operating when toxicity levels are dangerously in excess of limits.
- C.** Closing down the reserve when toxicity levels are genuinely in excess of allowed limits.
- D.** Keeping the reserve open and operating when toxicity levels are completely safe and within limits.
- E.** Closing the reserve entirely due to unrelated infrastructure noise level constraints.

Probability & Statistics — Variant 7

1. An outlier is best defined as an observation that:

- A. is seen more frequently than any other observations in the data set.
- B. is seen less frequently than the other observations in the data set.
- C. is always smaller than the rest of the observations in the data set.
- D. is always larger than the rest of the observations in the data set.
- E. is significantly and abnormally different from the other observations in the data set.

2. Management tracks the daily demand for internet bandwidth (Mbps) over several years. A cumulative probability chart compiled from the data follows a standard sigmoidal path where the 95th percentile corresponds precisely to a value of 850 Mbps.



How capacity or inventory should the center maintain daily to guarantee they fully satisfy demand on 95 percent of the days?

- A. At most 765 units
- B. At most 850 units
- C. Exactly 722 units
- D. At least 637 units
- E. At least 850 units

3. A large corporation has two separate primary operational divisions. The mean performance metric (or salary compensation baseline) of staff in Division 1 is 30 based on a cohort size of 80 employees. Division 2 features a mean baseline of 15 across a cohort size of 20 employees. What is the overall combined mean metric paid/allocated across all combined employees in this corporation?

- A. 15
- B. -977
- C. 22
- D. 27.0
- E. 30

4. A regional school district collects 75 years of weather closure datasets and establishes that the frequency distribution for days missed per year exhibits a pronounced right-hand skew (positive skewness) with an isolated far-right extreme outlier. Which of the following parameters should be chosen as the most accurate, robust measure to describe the center of this distribution?

- A. Mean, because it acts as an unbiased estimator for all linear samples.
- B. Median, because it remains resistant to extreme values and skewness.
- C. IQR, because it eliminates outliers and accounts for the central 50 percent.
- D. First quartile, because the distribution is left-skewed.
- E. Standard deviation, because it represents structural spread without bias.

5. The probability of an operational infrastructure failure incident occurring on any given business day depends strongly on environmental conditions. If weather systems remain dry, there is a 1.0% chance of an incident. If weather conditions shift to wet, the probability escalates to 3.0%. Today, the local meteorological station forecasts a 50% probability of wet conditions. What is the total probability that an operational incident occurs today?

- A. 0.0004
- B. 0.0016
- C. 0.0020
- D. 0.0200
- E. 0.0060

6. An institutional recruitment agency tracks retention rates carefully. Historical logs show that precisely 4% of all placed contractual candidates leave their positions by the end of their first year, and only 2% of those who successfully complete the first year leave during the subsequent second year. If an employer onboarded 100 workers via this firm at the start of Year 1, and onboarded another 100 workers at the start of Year 2, how many total workers are statistically expected to remain active on the job at the exact end of the second year?

- A. 171
- B. 180
- C. 186
- D. 190
- E. 199

7. The nutritional caloric value of a specific commercial item is normally distributed with an expected mean of 25 calories and a standard deviation of 2 calories. What percentage of these items contain more than 28.5 calories?

- A. 66.8%
- B. 47.7%
- C. 43.3%
- D. 6.68%
- E. 3.34%

8. A global distributor utilizes standard protective shipping units. The internal product contents packed per unit exhibit a mean weight of 25 lbs with a standard deviation of 6.0 lbs. The physical container itself has a standard deviation of 4.0 lbs, and the internal stabilization material has a standard deviation of 1.0 lbs. Assuming all component weights are independent, what is the exact standard deviation of the total packed shipping units?

- A. 28.000 pounds
- B. 5.290 pounds
- C. 4.085 pounds
- D. 2.400 pounds
- E. 7.280 pounds

9. A lead social researcher wants to estimate the true population mean annual income per family within a metropolitan school zone. Following a rigorous simple random sample, the researcher establishes a 95 percent confidence interval for the population mean spanning from \$35,095 to \$45,005. Which statement provides the most accurate statistical interpretation of this confidence interval?

- A. 95 percent of all families in this school zone have an annual income that falls within the interval limits.
- B. There is a 95% probability that the families of all students combined earn exactly between \$35,095 and \$45,005.
- C. If we were to take another sample of identical size and calculate a 95% confidence interval, there is a 95% chance that the new interval bounds will exactly match (\$35,095, \$45,005).
- D. There is a 95% probability that the mean of a subsequent sample drawn with matching sample size will fall between \$35,095 and \$45,005.
- E. In long-run repeated sampling, 95% of intervals generated using this exact methodology will successfully encompass the true fixed population mean income.

10. The municipal Department of Environmental Health tests soil toxicity across a local reserve. If the contamination level exceeds regulatory thresholds, the reserve is ordered to close immediately. The null hypothesis states H_0 : "The reserve is safe and non-toxic". Which option describes a final operational decision resulting directly from a Type I error?

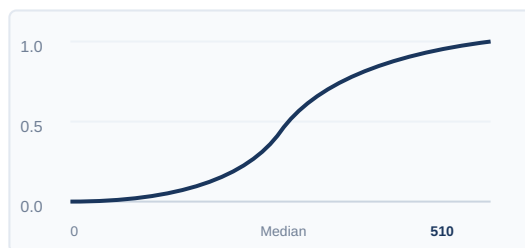
- A. Closing down the reserve when the toxicity levels are actually within regulatory safety limits (false alarm).
- B. Keeping the reserve open and operating when toxicity levels are dangerously in excess of limits.
- C. Closing down the reserve when toxicity levels are genuinely in excess of allowed limits.
- D. Keeping the reserve open and operating when toxicity levels are completely safe and within limits.
- E. Closing the reserve entirely due to unrelated infrastructure noise level constraints.

Probability & Statistics — Variant 8

1. An outlier is best defined as an observation that:

- A. is seen more frequently than any other observations in the data set.
- B. is seen less frequently than the other observations in the data set.
- C. is always smaller than the rest of the observations in the data set.
- D. is always larger than the rest of the observations in the data set.
- E. is significantly and abnormally different from the other observations in the data set.

2. Management tracks the daily demand for supermarket customers over several years. A cumulative probability chart compiled from the data follows a standard sigmoidal path where the 95th percentile corresponds precisely to a value of 510 Customers.



How capacity or inventory should the center maintain daily to guarantee they fully satisfy demand on 95 percent of the days?

- A. At most 459 units
- B. At most 510 units
- C. Exactly 433 units
- D. At least 382 units
- E. At least 510 units

3. A large corporation has two separate primary operational divisions. The mean performance metric (or salary compensation baseline) of staff in Division 1 is 15 based on a cohort size of 120 employees. Division 2 features a mean baseline of 18 across a cohort size of 80 employees. What is the overall combined mean metric paid/allocated across all combined employees in this corporation?

- A. 18
- B. -983
- C. 16
- D. 16.2
- E. 15

4. A regional school district collects 75 years of weather closure datasets and establishes that the frequency distribution for days missed per year exhibits a pronounced right-hand skew (positive skewness) with an isolated far-right extreme outlier. Which of the following parameters should be chosen as the most accurate, robust measure to describe the center of this distribution?

- A. Mean, because it acts as an unbiased estimator for all linear samples.
- B. Median, because it remains resistant to extreme values and skewness.
- C. IQR, because it eliminates outliers and accounts for the central 50 percent.
- D. First quartile, because the distribution is left-skewed.
- E. Standard deviation, because it represents structural spread without bias.

5. The probability of an operational infrastructure failure incident occurring on any given business day depends strongly on environmental conditions. If weather systems remain dry, there is a 3.0% chance of an incident. If weather conditions shift to wet, the probability escalates to 10.0%. Today, the local meteorological station forecasts a 5% probability of wet conditions. What is the total probability that an operational incident occurs today?

- A. 0.0004
- B. 0.0016
- C. 0.0020
- D. 0.0335
- E. 0.0060

6. An institutional recruitment agency tracks retention rates carefully. Historical logs show that precisely 10% of all placed contractual candidates leave their positions by the end of their first year, and only 4% of those who successfully complete the first year leave during the subsequent second year. If an employer onboarded 300 workers via this firm at the start of Year 1, and onboarded another 300 workers at the start of Year 2, how many total workers are statistically expected to remain active on the job at the exact end of the second year?

- A. 476
- B. 502
- C. 518
- D. 529
- E. 555

7. The nutritional caloric value of a specific commercial item is normally distributed with an expected mean of 70 calories and a standard deviation of 8 calories. What percentage of these items contain more than 82 calories?

- A. 66.8%
- B. 47.7%
- C. 43.3%
- D. 6.68%
- E. 3.34%

8. A global distributor utilizes standard protective shipping units. The internal product contents packed per unit exhibit a mean weight of 25 lbs with a standard deviation of 5.0 lbs. The physical container itself has a standard deviation of 2.0 lbs, and the internal stabilization material has a standard deviation of 0.5 lbs. Assuming all component weights are independent, what is the exact standard deviation of the total packed shipping units?

- A. 28.000 pounds
- B. 5.290 pounds
- C. 4.085 pounds
- D. 2.400 pounds
- E. 5.408 pounds

9. A lead social researcher wants to estimate the true population mean annual income per family within a metropolitan school zone. Following a rigorous simple random sample, the researcher establishes a 95 percent confidence interval for the population mean spanning from \$35,095 to \$45,005. Which statement provides the most accurate statistical interpretation of this confidence interval?

- A. 95 percent of all families in this school zone have an annual income that falls within the interval limits.
- B. There is a 95% probability that the families of all students combined earn exactly between \$35,095 and \$45,005.
- C. If we were to take another sample of identical size and calculate a 95% confidence interval, there is a 95% chance that the new interval bounds will exactly match (\$35,095, \$45,005).
- D. There is a 95% probability that the mean of a subsequent sample drawn with matching sample size will fall between \$35,095 and \$45,005.
- E. In long-run repeated sampling, 95% of intervals generated using this exact methodology will successfully encompass the true fixed population mean income.

10. The municipal Department of Environmental Health tests soil toxicity across a local reserve. If the contamination level exceeds regulatory thresholds, the reserve is ordered to close immediately. The null hypothesis states H_0 : "The reserve is safe and non-toxic". Which option describes a final operational decision resulting directly from a Type I error?

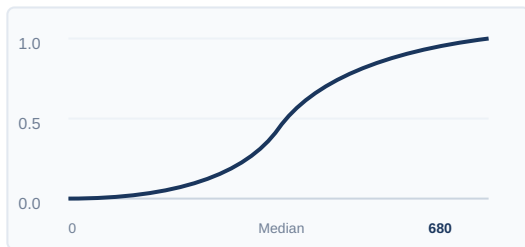
- A.** Closing down the reserve when the toxicity levels are actually within regulatory safety limits (false alarm).
- B.** Keeping the reserve open and operating when toxicity levels are dangerously in excess of limits.
- C.** Closing down the reserve when toxicity levels are genuinely in excess of allowed limits.
- D.** Keeping the reserve open and operating when toxicity levels are completely safe and within limits.
- E.** Closing the reserve entirely due to unrelated infrastructure noise level constraints.

Probability & Statistics — Variant 9

1. An outlier is best defined as an observation that:

- A. is seen more frequently than any other observations in the data set.
- B. is seen less frequently than the other observations in the data set.
- C. is always smaller than the rest of the observations in the data set.
- D. is always larger than the rest of the observations in the data set.
- E. is significantly and abnormally different from the other observations in the data set.

2. Management tracks the daily demand for courier packages delivered over several years. A cumulative probability chart compiled from the data follows a standard sigmoidal path where the 95th percentile corresponds precisely to a value of 680 Packages.



How capacity or inventory should the center maintain daily to guarantee they fully satisfy demand on 95 percent of the days?

- A. At most 612 units
- B. At most 680 units
- C. Exactly 578 units
- D. At least 510 units
- E. At least 680 units

3. A large corporation has two separate primary operational divisions. The mean performance metric (or salary compensation baseline) of staff in Division 1 is 82 based on a cohort size of 45 employees. Division 2 features a mean baseline of 94 across a cohort size of 15 employees. What is the overall combined mean metric paid/allocated across all combined employees in this corporation?

- A. 94
- B. -912
- C. 88
- D. 85.0
- E. 82

4. A regional school district collects 75 years of weather closure datasets and establishes that the frequency distribution for days missed per year exhibits a pronounced right-hand skew (positive skewness) with an isolated far-right extreme outlier. Which of the following parameters should be chosen as the most accurate, robust measure to describe the center of this distribution?

- A.** Mean, because it acts as an unbiased estimator for all linear samples.
- B.** Median, because it remains resistant to extreme values and skewness.
- C.** IQR, because it eliminates outliers and accounts for the central 50 percent.
- D.** First quartile, because the distribution is left-skewed.
- E.** Standard deviation, because it represents structural spread without bias.

5. The probability of an operational infrastructure failure incident occurring on any given business day depends strongly on environmental conditions. If weather systems remain dry, there is a 2.0% chance of an incident. If weather conditions shift to wet, the probability escalates to 4.0%. Today, the local meteorological station forecasts a 35% probability of wet conditions. What is the total probability that an operational incident occurs today?

- A.** 0.0004
- B.** 0.0016
- C.** 0.0020
- D.** 0.0270
- E.** 0.0060

6. An institutional recruitment agency tracks retention rates carefully. Historical logs show that precisely 12% of all placed contractual candidates leave their positions by the end of their first year, and only 5% of those who successfully complete the first year leave during the subsequent second year. If an employer onboarded 500 workers via this firm at the start of Year 1, and onboarded another 500 workers at the start of Year 2, how many total workers are statistically expected to remain active on the job at the exact end of the second year?

- A.** 772
- B.** 815
- C.** 840
- D.** 858
- E.** 900

7. The nutritional caloric value of a specific commercial item is normally distributed with an expected mean of 150 calories and a standard deviation of 14 calories. What percentage of these items contain more than 171 calories?

- A. 66.8%**
- B. 47.7%**
- C. 43.3%**
- D. 6.68%**
- E. 3.34%**

8. A global distributor utilizes standard protective shipping units. The internal product contents packed per unit exhibit a mean weight of 25 lbs with a standard deviation of 1.5 lbs. The physical container itself has a standard deviation of 0.2 lbs, and the internal stabilization material has a standard deviation of 0.4 lbs. Assuming all component weights are independent, what is the exact standard deviation of the total packed shipping units?

- A. 28.000 pounds**
- B. 5.290 pounds**
- C. 4.085 pounds**
- D. 2.400 pounds**
- E. 1.565 pounds**

9. A lead social researcher wants to estimate the true population mean annual income per family within a metropolitan school zone. Following a rigorous simple random sample, the researcher establishes a 95 percent confidence interval for the population mean spanning from \$35,095 to \$45,005. Which statement provides the most accurate statistical interpretation of this confidence interval?

- A. 95 percent of all families in this school zone have an annual income that falls within the interval limits.**
- B. There is a 95% probability that the families of all students combined earn exactly between \$35,095 and \$45,005.**
- C. If we were to take another sample of identical size and calculate a 95% confidence interval, there is a 95% chance that the new interval bounds will exactly match (\$35,095, \$45,005).**
- D. There is a 95% probability that the mean of a subsequent sample drawn with matching sample size will fall between \$35,095 and \$45,005.**
- E. In long-run repeated sampling, 95% of intervals generated using this exact methodology will successfully encompass the true fixed population mean income.**

10. The municipal Department of Environmental Health tests soil toxicity across a local reserve. If the contamination level exceeds regulatory thresholds, the reserve is ordered to close immediately. The null hypothesis states H_0 : "The reserve is safe and non-toxic". Which option describes a final operational decision resulting directly from a Type I error?

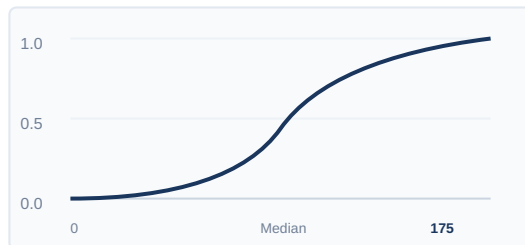
- A.** Closing down the reserve when the toxicity levels are actually within regulatory safety limits (false alarm).
- B.** Keeping the reserve open and operating when toxicity levels are dangerously in excess of limits.
- C.** Closing down the reserve when toxicity levels are genuinely in excess of allowed limits.
- D.** Keeping the reserve open and operating when toxicity levels are completely safe and within limits.
- E.** Closing the reserve entirely due to unrelated infrastructure noise level constraints.

Probability & Statistics — Variant 10

1. An outlier is best defined as an observation that:

- A. is seen more frequently than any other observations in the data set.
- B. is seen less frequently than the other observations in the data set.
- C. is always smaller than the rest of the observations in the data set.
- D. is always larger than the rest of the observations in the data set.
- E. is significantly and abnormally different from the other observations in the data set.

2. Management tracks the daily demand for car rentals per day over several years. A cumulative probability chart compiled from the data follows a standard sigmoidal path where the 95th percentile corresponds precisely to a value of 175 Cars.



How capacity or inventory should the center maintain daily to guarantee they fully satisfy demand on 95 percent of the days?

- A. At most 157 units
- B. At most 175 units
- C. Exactly 148 units
- D. At least 131 units
- E. At least 175 units

3. A large corporation has two separate primary operational divisions. The mean performance metric (or salary compensation baseline) of staff in Division 1 is 8 based on a cohort size of 12 employees. Division 2 features a mean baseline of 14 across a cohort size of 4 employees. What is the overall combined mean metric paid/allocated across all combined employees in this corporation?

- A. 14
- B. -989
- C. 11
- D. 9.5
- E. 8

4. A regional school district collects 75 years of weather closure datasets and establishes that the frequency distribution for days missed per year exhibits a pronounced right-hand skew (positive skewness) with an isolated far-right extreme outlier. Which of the following parameters should be chosen as the most accurate, robust measure to describe the center of this distribution?

- A.** Mean, because it acts as an unbiased estimator for all linear samples.
- B.** Median, because it remains resistant to extreme values and skewness.
- C.** IQR, because it eliminates outliers and accounts for the central 50 percent.
- D.** First quartile, because the distribution is left-skewed.
- E.** Standard deviation, because it represents structural spread without bias.

5. The probability of an operational infrastructure failure incident occurring on any given business day depends strongly on environmental conditions. If weather systems remain dry, there is a 1.0% chance of an incident. If weather conditions shift to wet, the probability escalates to 12.0%. Today, the local meteorological station forecasts a 8% probability of wet conditions. What is the total probability that an operational incident occurs today?

- A.** 0.0004
- B.** 0.0016
- C.** 0.0020
- D.** 0.0188
- E.** 0.0060

6. An institutional recruitment agency tracks retention rates carefully. Historical logs show that precisely 8% of all placed contractual candidates leave their positions by the end of their first year, and only 4% of those who successfully complete the first year leave during the subsequent second year. If an employer onboarded 50 workers via this firm at the start of Year 1, and onboarded another 50 workers at the start of Year 2, how many total workers are statistically expected to remain active on the job at the exact end of the second year?

- A.** 82
- B.** 87
- C.** 90
- D.** 92
- E.** 96

7. The nutritional caloric value of a specific commercial item is normally distributed with an expected mean of 1000 calories and a standard deviation of 80 calories. What percentage of these items contain more than 1120 calories?

- A.** 66.8%
- B.** 47.7%
- C.** 43.3%
- D.** 6.68%
- E.** 3.34%

8. A global distributor utilizes standard protective shipping units. The internal product contents packed per unit exhibit a mean weight of 25 lbs with a standard deviation of 1.0 lbs. The physical container itself has a standard deviation of 0.6 lbs, and the internal stabilization material has a standard deviation of 0.2 lbs. Assuming all component weights are independent, what is the exact standard deviation of the total packed shipping units?

- A.** 28.000 pounds
- B.** 5.290 pounds
- C.** 4.085 pounds
- D.** 2.400 pounds
- E.** 1.183 pounds

9. A lead social researcher wants to estimate the true population mean annual income per family within a metropolitan school zone. Following a rigorous simple random sample, the researcher establishes a 95 percent confidence interval for the population mean spanning from \$35,095 to \$45,005. Which statement provides the most accurate statistical interpretation of this confidence interval?

- A.** 95 percent of all families in this school zone have an annual income that falls within the interval limits.
- B.** There is a 95% probability that the families of all students combined earn exactly between \$35,095 and \$45,005.
- C.** If we were to take another sample of identical size and calculate a 95% confidence interval, there is a 95% chance that the new interval bounds will exactly match (\$35,095, \$45,005).
- D.** There is a 95% probability that the mean of a subsequent sample drawn with matching sample size will fall between \$35,095 and \$45,005.
- E.** In long-run repeated sampling, 95% of intervals generated using this exact methodology will successfully encompass the true fixed population mean income.

10. The municipal Department of Environmental Health tests soil toxicity across a local reserve. If the contamination level exceeds regulatory thresholds, the reserve is ordered to close immediately. The null hypothesis states H_0 : "The reserve is safe and non-toxic". Which option describes a final operational decision resulting directly from a Type I error?

- A.** Closing down the reserve when the toxicity levels are actually within regulatory safety limits (false alarm).
- B.** Keeping the reserve open and operating when toxicity levels are dangerously in excess of limits.
- C.** Closing down the reserve when toxicity levels are genuinely in excess of allowed limits.
- D.** Keeping the reserve open and operating when toxicity levels are completely safe and within limits.
- E.** Closing the reserve entirely due to unrelated infrastructure noise level constraints.

Master Answer Key (Variants 1–10)

This master key lists the correct option letter for each of the 10 parallel multiple choice questions across all variants generated in this book.

Question No.	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10
Question 1	E	E	E	E	E	E	E	E	E	E
Question 2	E	E	E	E	E	E	E	E	E	E
Question 3	D	D	D	D	D	D	D	D	D	D
Question 4	B	B	B	B	B	B	B	B	B	B
Question 5	D	D	D	D	D	D	D	D	D	D
Question 6	D	D	D	D	D	D	D	D	D	D
Question 7	D	D	D	D	D	D	D	D	D	D
Question 8	E	E	E	E	E	E	E	E	E	E
Question 9	E	E	E	E	E	E	E	E	E	E
Question 10	A	A	A	A	A	A	A	A	A	A